

Data written to the working file.
15 variables and 22950 cases written.
Variable: component_type Type: String Format : A10
Variable: contrast_deutanopia Type: Number Format : F7.4
Variable: contrast_loss_deutanopia Type: Number Format : F7.4
Variable: contrast_loss_protanopia Type: Number Format : F7.4
Variable: contrast_loss_tritanopia Type: Number Format : F7.4
Variable: contrast_normal Type: Number Format : F7.4
Variable: contrast_protanopia Type: Number Format : F7.4
Variable: contrast_tritanopia Type: Number Format : F7.4
Variable: pass_deutanopia Type: Number Format : F1
Variable: pass_normal Type: Number Format : F1
Variable: pass_protanopia Type: Number Format : F1
Variable: pass_tritanopia Type: Number Format : F1
Variable: site Type: String Format : A43
Variable: tag Type: String Format : A43
Variable: wcag_threshold Type: Number Format : F3.1

Substitute the following to build syntax for these data.

```
/VARIABLES=
component_type A10
contrast_deutanopia F7.4
contrast_loss_deutanopia F7.4
contrast_loss_protanopia F7.4
contrast_loss_tritanopia F7.4
contrast_normal F7.4
contrast_protanopia F7.4
contrast_tritanopia F7.4
pass_deutanopia F1
pass_normal F1
pass_protanopia F1
pass_tritanopia F1
site A43
tag A43
wcag_threshold F3.1
```

Descriptives

Notes

Output Created		20-APR-2026 18:42:21
Comments		
Input	Data	/Users/lukevassallo/Dev/6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=contrast_pro tanopia contrast_tritanopia contrast_deutanopia /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.02
	Elapsed Time	00:00:00.00

[DataSet1]

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
contrast_protanopia	22950	1.0000	20.8220	11.144993	6.0322317
contrast_tritanopia	22950	1.0000	20.8220	11.064428	6.0876712
contrast_deutanopia	22950	1.0000	20.8220	11.030277	6.1217345
Valid N (listwise)	22950				

Descriptives

Notes

Output Created		20-APR-2026 18:42:55
Comments		
Input	Data	/Users/lukevassallo/Dev /6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User defined missing values are treated as missing.
	Cases Used	All non-missing data are used.
Syntax		DESCRIPTIVES VARIABLES=pass_protanopia pass_deutanopia pass_tritanopia /STATISTICS=MEAN STDDEV MIN MAX.
Resources	Processor Time	00:00:00.03
	Elapsed Time	00:00:00.00

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
pass_protanopia	22950	0	1	.85	.354
pass_deutanopia	22950	0	1	.85	.359
pass_tritanopia	22950	0	1	.85	.354
Valid N (listwise)	22950				

component_type into component_type_num
Old Value New Value Value Label

alert 1 alert
button 2 button
card 3 card
form 4 form
header 5 header
input 6 input
link 7 link
navigation 8 navigation
other 9 other
text 10 text

Oneway

Notes

Output Created		20-APR-2026 18:46:28
Comments		
Input	Data	/Users/lukevassallo/Dev /6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY contrast_protanopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.00

ANOVA

contrast_protanopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	5284.375	9	587.153	16.232	<.001
Within Groups	829779.686	22940	36.172		
Total	835064.061	22949			

ANOVA Effect Sizes^a

			95% Confidence Interval	
		Point Estimate	Lower	Upper
contrast_protanopia	Eta-squared	.006	.004	.008
	Epsilon-squared	.006	.004	.008
	Omega-squared Fixed-effect	.006	.004	.008
	Omega-squared Random-effect	.001	.000	.001

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	6.2613346	2.6902374	.371
	card	4.6461067	2.7163318	.790
	form	5.5091709	3.2438692	.797
	header	6.7445286	2.8656522	.354
	input	2.7893550	4.0345114	1.000
	link	7.4743252	2.6924792	.145
	navigation	4.3449505	2.7282656	.852
	other	6.6743301	2.6908718	.279
	text	6.1643122	2.6923423	.396
button	alert	-6.2613346	2.6902374	.371
	card	-1.6152279*	.3835883	.001
	form	-.7521637	1.8142131	1.000
	header	.4831941	.9902747	1.000
	input	-3.4719796	3.0076509	.979
	link	1.2129906*	.1346335	<.001
	navigation	-1.9163841*	.4605591	.001
	other	.4129955*	.0973272	<.001
	text	-.0970224	.1318674	.999
card	alert	-4.6461067	2.7163318	.790
	button	1.6152279*	.3835883	.001
	form	.8630642	1.8526873	1.000
	header	2.0984220	1.0591152	.612
	input	-1.8567517	3.0310139	1.000
	link	2.8282185*	.3990078	<.001
	navigation	-.3011562	.5943027	1.000
	other	2.0282234*	.3880124	<.001
	text	1.5182055*	.3980830	.005

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
alert	button	-2.250568	14.773238
	card	-3.948359	13.240572
	form	-4.754422	15.772764
	header	-2.322386	15.811444
	input	-9.975827	15.554537
	link	-1.044671	15.993321
	navigation	-4.287274	12.977175
	other	-1.839580	15.188240
	text	-2.354251	14.682875
button	alert	-14.773238	2.250568
	card	-2.828900	-.401556
	form	-6.492329	4.988001
	header	-2.650032	3.616420
	input	-12.988178	6.044219
	link	.787011	1.638971
	navigation	-3.373592	-.459177
	other	.105052	.720939
	text	-.514251	.320206
card	alert	-13.240572	3.948359
	button	.401556	2.828900
	form	-4.998833	6.724961
	header	-1.252615	5.449459
	input	-11.446871	7.733367
	link	1.565759	4.090678
	navigation	-2.181528	1.579216
	other	.800553	3.255894
	text	.258672	2.777739

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
form	alert	-5.5091709	3.2438692	.797
	button	.7521637	1.8142131	1.000
	card	-.8630642	1.8526873	1.000
	header	1.2353577	2.0654187	1.000
	input	-2.7198159	3.5115915	.999
	link	1.9651543	1.8175359	.987
	navigation	-1.1642204	1.8701404	1.000
	other	1.1651591	1.8151537	1.000
	text	.6551413	1.8173331	1.000
header	alert	-6.7445286	2.8656522	.354
	button	-.4831941	.9902747	1.000
	card	-2.0984220	1.0591152	.612
	form	-1.2353577	2.0654187	1.000
	input	-3.9551736	3.1655251	.964
	link	.7297966	.9963490	.999
	navigation	-2.3995781	1.0893575	.455
	other	-.0701986	.9919968	1.000
	text	-.5802165	.9959790	1.000
input	alert	-2.7893550	4.0345114	1.000
	button	3.4719796	3.0076509	.979
	card	1.8567517	3.0310139	1.000
	form	2.7198159	3.5115915	.999
	header	3.9551736	3.1655251	.964
	link	4.6849702	3.0096564	.869
	navigation	1.5555955	3.0417133	1.000
	other	3.8849751	3.0082184	.956
	text	3.3749572	3.0095339	.983
link	alert	-7.4743252	2.6924792	.145
	button	-1.2129906*	.1346335	<.001
	card	-2.8282185*	.3990078	<.001
	form	-1.9651543	1.8175359	.987
	header	-.7297966	.9963490	.999
	input	-4.6849702	3.0096564	.869
	navigation	-3.1293747*	.4734785	<.001
	other	-.7999951*	.1467646	<.001
	text	-1.3100130*	.1716283	<.001
navigation	alert	-4.3449505	2.7282656	.852
	button	1.9163841*	.4605591	.001
	card	.3011562	.5943027	1.000
	form	1.1642204	1.8701404	1.000
	header	2.3995781	1.0893575	.455

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
form	alert	-15.772764	4.754422
	button	-4.988001	6.492329
	card	-6.724961	4.998833
	header	-5.299621	7.770336
	input	-13.830481	8.390849
	link	-3.785524	7.715832
	navigation	-7.081339	4.752898
	other	-4.577982	6.908300
	text	-5.094895	6.405178
header	alert	-15.811444	2.322386
	button	-3.616420	2.650032
	card	-5.449459	1.252615
	form	-7.770336	5.299621
	input	-13.970886	6.060538
	link	-2.422649	3.882242
	navigation	-5.846302	1.047146
	other	-3.208874	3.068476
	text	-3.731491	2.571058
input	alert	-15.554537	9.975827
	button	-6.044219	12.988178
	card	-7.733367	11.446871
	form	-8.390849	13.830481
	header	-6.060538	13.970886
	link	-4.837574	14.207514
	navigation	-8.068376	11.179567
	other	-5.633019	13.402969
	text	-6.147199	12.897113
link	alert	-15.993321	1.044671
	button	-1.638971	-.787011
	card	-4.090678	-1.565759
	form	-7.715832	3.785524
	header	-3.882242	2.422649
	input	-14.207514	4.837574
	navigation	-4.627459	-1.631290
	other	-1.264358	-.335632
	text	-1.853045	-.766981
navigation	alert	-12.977175	4.287274
	button	.459177	3.373592
	card	-1.579216	2.181528
	form	-4.752898	7.081339
	header	-1.047146	5.846302

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	input	-1.5555955	3.0417133	1.000
	link	3.1293747*	.4734785	<.001
	other	2.3293795*	.4642503	<.001
	text	1.8193617*	.4726994	.005
other	alert	-6.6743301	2.6908718	.279
	button	-.4129955*	.0973272	<.001
	card	-2.0282234*	.3880124	<.001
	form	-1.1651591	1.8151537	1.000
	header	.0701986	.9919968	1.000
	input	-3.8849751	3.0082184	.956
	link	.7999951*	.1467646	<.001
	navigation	-2.3293795*	.4642503	<.001
	text	-.5100179*	.1442314	.015
	other			
text	alert	-6.1643122	2.6923423	.396
	button	.0970224	.1318674	.999
	card	-1.5182055*	.3980830	.005
	form	-.6551413	1.8173331	1.000
	header	.5802165	.9959790	1.000
	input	-3.3749572	3.0095339	.983
	link	1.3100130*	.1716283	<.001
	navigation	-1.8193617*	.4726994	.005
	other	.5100179*	.1442314	.015
	text			

Multiple Comparisons

Dependent Variable: contrast_protanopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
	input	-11.179567	8.068376
	link	1.631290	4.627459
	other	.860493	3.798266
	text	.323742	3.314981
other	alert	-15.188240	1.839580
	button	-.720939	-.105052
	card	-3.255894	-.800553
	form	-6.908300	4.577982
	header	-3.068476	3.208874
	input	-13.402969	5.633019
	link	.335632	1.264358
	navigation	-3.798266	-.860493
	text	-.966366	-.053670
text	alert	-14.682875	2.354251
	button	-.320206	.514251
	card	-2.777739	-.258672
	form	-6.405178	5.094895
	header	-2.571058	3.731491
	input	-12.897113	6.147199
	link	.766981	1.853045
	navigation	-3.314981	-.323742
	other	.053670	.966366

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

contrast_protanopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05	
		1	2
link	2396	10.115355	
header	37	10.845151	
other	5614	10.915350	
button	11940	11.328345	11.328345
text	2519	11.425368	11.425368
form	11	12.080509	12.080509
card	251	12.943573	12.943573
navigation	173	13.244729	13.244729
input	4	14.800325	14.800325
alert	5		17.589680
Sig.		.396	.068

Means for groups in homogeneous subsets are displayed.

- Uses Harmonic Mean Sample Size = 17.278.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Oneway

Notes

Output Created		20-APR-2026 18:47:15
Comments		
Input	Data	/Users/lukevassallo/Dev /6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.

Notes

Syntax		ONEWAY contrast_deutanopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).
Resources	Processor Time	00:00:00.07
	Elapsed Time	00:00:00.00

ANOVA

contrast_deutanopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	5577.288	9	619.699	16.637	<.001
Within Groups	854451.022	22940	37.247		
Total	860028.310	22949			

ANOVA Effect Sizes^a

		Point Estimate	95% Confidence Interval	
			Lower	Upper
contrast_deutanopia	Eta-squared	.006	.004	.008
	Epsilon-squared	.006	.004	.008
	Omega-squared Fixed-effect	.006	.004	.008
	Omega-squared Random-effect	.001	.000	.001

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	6.3579064	2.7299380	.370
	card	4.7798696	2.7564175	.776
	form	5.3933709	3.2917400	.829
	header	6.7287422	2.9079415	.380
	input	2.8864550	4.0940499	.999
	link	7.5982212	2.7322130	.143
	navigation	4.3048794	2.7685274	.870
	other	6.7872404	2.7305818	.276
	text	6.2373943	2.7320741	.400
button	alert	-6.3579064	2.7299380	.370
	card	-1.5780367*	.3892490	.002
	form	-.9645355	1.8409860	1.000
	header	.3708358	1.0048885	1.000
	input	-3.4714514	3.0520358	.981
	link	1.2403148*	.1366203	<.001
	navigation	-2.0530269*	.4673557	<.001
	other	.4293340*	.0987635	<.001
	text	-.1205121	.1338134	.996
card	alert	-4.7798696	2.7564175	.776
	button	1.5780367*	.3892490	.002
	form	.6135013	1.8800280	1.000
	header	1.9488725	1.0747449	.727
	input	-1.8934146	3.0757435	1.000
	link	2.8183516*	.4048961	<.001
	navigation	-.4749902	.6030730	.999
	other	2.0073707*	.3937384	<.001
	text	1.4575246*	.4039576	.012
form	alert	-5.3933709	3.2917400	.829
	button	.9645355	1.8409860	1.000
	card	-.6135013	1.8800280	1.000
	header	1.3353713	2.0958987	1.000
	input	-2.5069159	3.5634131	1.000
	link	2.2048503	1.8443578	.973
	navigation	-1.0884915	1.8977386	1.000
	other	1.3938695	1.8419405	.999
	text	.8440234	1.8441520	1.000
header	alert	-6.7287422	2.9079415	.380
	button	-.3708358	1.0048885	1.000
	card	-1.9488725	1.0747449	.727
	form	-1.3353713	2.0958987	1.000
	input	-3.8422872	3.2122397	.973

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
alert	button	-2.279609	14.995422
	card	-3.941427	13.501166
	form	-5.021685	15.808427
	header	-2.471976	15.929460
	input	-10.067107	15.840017
	link	-1.046493	16.242935
	navigation	-4.454733	13.064492
	other	-1.852312	15.426793
	text	-2.406880	14.881669
button	alert	-14.995422	2.279609
	card	-2.809619	-.346454
	form	-6.789410	4.860339
	header	-2.808628	3.550300
	input	-13.128083	6.185181
	link	.808049	1.672581
	navigation	-3.531739	-.574315
	other	.116847	.741822
	text	-.543897	.302873
card	alert	-13.501166	3.941427
	button	.346454	2.809619
	form	-5.334902	6.561904
	header	-1.451617	5.349362
	input	-11.625058	7.838229
	link	1.537262	4.099442
	navigation	-2.383111	1.433131
	other	.761583	3.253158
	text	.179404	2.735645
form	alert	-15.808427	5.021685
	button	-4.860339	6.789410
	card	-6.561904	5.334902
	header	-5.296046	7.966789
	input	-13.781544	8.767712
	link	-3.630692	8.040393
	navigation	-7.092931	4.915948
	other	-4.434025	7.221764
	text	-4.990868	6.678915
header	alert	-15.929460	2.471976
	button	-3.550300	2.808628
	card	-5.349362	1.451617
	form	-7.966789	5.296046
	input	-14.005804	6.321230

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	link	.8694790	1.0110524	.998
	navigation	-2.4238627	1.1054335	.462
	other	.0584982	1.0066360	1.000
	text	-.4913479	1.0106769	1.000
input	alert	-2.8864550	4.0940499	.999
	button	3.4714514	3.0520358	.981
	card	1.8934146	3.0757435	1.000
	form	2.5069159	3.5634131	1.000
	header	3.8422872	3.2122397	.973
	link	4.7117662	3.0540708	.875
	navigation	1.4184244	3.0866008	1.000
	other	3.9007854	3.0526116	.959
	text	3.3509393	3.0539465	.985
link	alert	-7.5982212	2.7322130	.143
	button	-1.2403148*	.1366203	<.001
	card	-2.8183516*	.4048961	<.001
	form	-2.2048503	1.8443578	.973
	header	-.8694790	1.0110524	.998
	input	-4.7117662	3.0540708	.875
	navigation	-3.2933418*	.4804658	<.001
	other	-.8109808*	.1489305	<.001
	text	-1.3608269*	.1741611	<.001
navigation	alert	-4.3048794	2.7685274	.870
	button	2.0530269*	.4673557	<.001
	card	.4749902	.6030730	.999
	form	1.0884915	1.8977386	1.000
	header	2.4238627	1.1054335	.462
	input	-1.4184244	3.0866008	1.000
	link	3.2933418*	.4804658	<.001
	other	2.4823610*	.4711014	<.001
	text	1.9325149*	.4796752	.002
other	alert	-6.7872404	2.7305818	.276
	button	-.4293340*	.0987635	<.001
	card	-2.0073707*	.3937384	<.001
	form	-1.3938695	1.8419405	.999
	header	-.0584982	1.0066360	1.000
	input	-3.9007854	3.0526116	.959
	link	.8109808*	.1489305	<.001
	navigation	-2.4823610*	.4711014	<.001
	text	-.5498461*	.1463599	.007

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
	link	-2.329488	4.068446
	navigation	-5.921451	1.073725
	other	-3.126495	3.243492
	text	-3.689127	2.706431
input	alert	-15.840017	10.067107
	button	-6.185181	13.128083
	card	-7.838229	11.625058
	form	-8.767712	13.781544
	header	-6.321230	14.005804
	link	-4.951305	14.374837
	navigation	-8.347571	11.184420
	other	-5.757669	13.559239
	text	-6.311738	13.013617
link	alert	-16.242935	1.046493
	button	-1.672581	-.808049
	card	-4.099442	-1.537262
	form	-8.040393	3.630692
	header	-4.068446	2.329488
	input	-14.374837	4.951305
	navigation	-4.813534	-1.773149
	other	-1.282196	-.339765
	text	-1.911872	-.809782
navigation	alert	-13.064492	4.454733
	button	.574315	3.531739
	card	-1.433131	2.383111
	form	-4.915948	7.092931
	header	-1.073725	5.921451
	input	-11.184420	8.347571
	link	1.773149	4.813534
	other	.991798	3.972924
	text	.414824	3.450206
other	alert	-15.426793	1.852312
	button	-.741822	-.116847
	card	-3.253158	-.761583
	form	-7.221764	4.434025
	header	-3.243492	3.126495
	input	-13.559239	5.757669
	link	.339765	1.282196
	navigation	-3.972924	-.991798
	text	-1.012928	-.086764

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
text	alert	-6.2373943	2.7320741	.400
	button	.1205121	.1338134	.996
	card	-1.4575246*	.4039576	.012
	form	-.8440234	1.8441520	1.000
	header	.4913479	1.0106769	1.000
	input	-3.3509393	3.0539465	.985
	link	1.3608269*	.1741611	<.001
	navigation	-1.9325149*	.4796752	.002
	other	.5498461*	.1463599	.007

Multiple Comparisons

Dependent Variable: contrast_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
text	alert	-14.881669	2.406880
	button	-.302873	.543897
	card	-2.735645	-.179404
	form	-6.678915	4.990868
	header	-2.706431	3.689127
	input	-13.013617	6.311738
	link	.809782	1.911872
	navigation	-3.450206	-.414824
	other	.086764	1.012928

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

contrast_deutanopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05	
		1	2
link	2396	9.976659	
other	5614	10.787640	
header	37	10.846138	
button	11940	11.216974	11.216974
text	2519	11.337486	11.337486
form	11	12.181509	12.181509
card	251	12.795010	12.795010
navigation	173	13.270001	13.270001
input	4	14.688425	14.688425
alert	5		17.574880
Sig.		.409	.068

Means for groups in homogeneous subsets are displayed.

- Uses Harmonic Mean Sample Size = 17.278.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Oneway

Notes

Output Created		20-APR-2026 18:47:50
Comments		
Input	Data	/Users/lukevassallo/Dev /6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.

Notes

Syntax	ONEWAY contrast_tritanopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).	
Resources	Processor Time	00:00:00.07
	Elapsed Time	00:00:00.00

ANOVA

contrast_tritanopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	5377.657	9	597.517	16.219	<.001
Within Groups	845106.338	22940	36.840		
Total	850483.995	22949			

ANOVA Effect Sizes^a

		Point Estimate	95% Confidence Interval	
			Lower	Upper
contrast_tritanopia	Eta-squared	.006	.004	.008
	Epsilon-squared	.006	.004	.008
	Omega-squared Fixed-effect	.006	.004	.008
	Omega-squared Random-effect	.001	.000	.001

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	6.3310970	2.7149691	.368
	card	4.7572999	2.7413033	.775
	form	5.4530527	3.2736905	.815
	header	6.7839692	2.8919964	.359
	input	2.8540300	4.0716012	1.000
	link	7.5482053	2.7172315	.144
	navigation	4.3279193	2.7533468	.862
	other	6.7567926	2.7156093	.275
	text	6.2303648	2.7170934	.393
button	alert	-6.3310970	2.7149691	.368
	card	-1.5737970*	.3871147	.002
	form	-.8780442	1.8308914	1.000
	header	.4528722	.9993784	1.000
	input	-3.4770670	3.0353006	.980
	link	1.2171083*	.1358712	<.001
	navigation	-2.0031777*	.4647930	<.001
	other	.4256956*	.0982220	<.001
	text	-.1007322	.1330797	.999
card	alert	-4.7572999	2.7413033	.775
	button	1.5737970*	.3871147	.002
	form	.6957528	1.8697193	1.000
	header	2.0266693	1.0688518	.672
	input	-1.9032699	3.0588784	1.000
	link	2.7909054*	.4026759	<.001
	navigation	-.4293806	.5997662	.999
	other	1.9994926*	.3915795	<.001
	text	1.4730649*	.4017426	.009
form	alert	-5.4530527	3.2736905	.815
	button	.8780442	1.8308914	1.000
	card	-.6957528	1.8697193	1.000
	header	1.3309165	2.0844063	1.000
	input	-2.5990227	3.5438739	.999
	link	2.0951526	1.8342447	.980
	navigation	-1.1251334	1.8873328	1.000
	other	1.3037398	1.8318406	.999
	text	.7773121	1.8340400	1.000
header	alert	-6.7839692	2.8919964	.359
	button	-.4528722	.9993784	1.000
	card	-2.0266693	1.0688518	.672
	form	-1.3309165	2.0844063	1.000
	input	-3.9299392	3.1946262	.968

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
alert	button	-2.259057	14.921251
	card	-3.916176	13.430775
	form	-4.904894	15.811000
	header	-2.366299	15.934237
	input	-10.028504	15.736564
	link	-1.049107	16.145518
	navigation	-4.383662	13.039500
	other	-1.835387	15.348972
	text	-2.366511	14.827240
button	alert	-14.921251	2.259057
	card	-2.798627	-.348967
	form	-6.670979	4.914891
	header	-2.709158	3.614903
	input	-13.080749	6.126615
	link	.787212	1.647004
	navigation	-3.473781	-.532574
	other	.114922	.736470
	text	-.521796	.320332
card	alert	-13.430775	3.916176
	button	.348967	2.798627
	form	-5.220034	6.611539
	header	-1.355175	5.408513
	input	-11.581552	7.775012
	link	1.516840	4.064971
	navigation	-2.327039	1.468278
	other	.760536	3.238449
	text	.201952	2.744177
form	alert	-15.811000	4.904894
	button	-4.914891	6.670979
	card	-6.611539	5.220034
	header	-5.264139	7.925972
	input	-13.811829	8.613784
	link	-3.708392	7.898697
	navigation	-7.096649	4.846382
	other	-4.492198	7.099678
	text	-5.025585	6.580209
header	alert	-15.934237	2.366299
	button	-3.614903	2.709158
	card	-5.408513	1.355175
	form	-7.925972	5.264139
	input	-14.037727	6.177848

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	link	.7642361	1.0055085	.999
	navigation	-2.4560499	1.0993721	.433
	other	-.0271766	1.0011164	1.000
	text	-.5536044	1.0051351	1.000
input	alert	-2.8540300	4.0716012	1.000
	button	3.4770670	3.0353006	.980
	card	1.9032699	3.0588784	1.000
	form	2.5990227	3.5438739	.999
	header	3.9299392	3.1946262	.968
	link	4.6941753	3.0373245	.874
	navigation	1.4738893	3.0696761	1.000
	other	3.9027626	3.0358733	.957
	text	3.3763348	3.0372009	.984
link	alert	-7.5482053	2.7172315	.144
	button	-1.2171083*	.1358712	<.001
	card	-2.7909054*	.4026759	<.001
	form	-2.0951526	1.8342447	.980
	header	-.7642361	1.0055085	.999
	input	-4.6941753	3.0373245	.874
	navigation	-3.2202860*	.4778313	<.001
	other	-.7914127*	.1481138	<.001
	text	-1.3178405*	.1732061	<.001
navigation	alert	-4.3279193	2.7533468	.862
	button	2.0031777*	.4647930	<.001
	card	.4293806	.5997662	.999
	form	1.1251334	1.8873328	1.000
	header	2.4560499	1.0993721	.433
	input	-1.4738893	3.0696761	1.000
	link	3.2202860*	.4778313	<.001
	other	2.4288733*	.4685182	<.001
	text	1.9024455*	.4770450	.003
other	alert	-6.7567926	2.7156093	.275
	button	-.4256956*	.0982220	<.001
	card	-1.9994926*	.3915795	<.001
	form	-1.3037398	1.8318406	.999
	header	.0271766	1.0011164	1.000
	input	-3.9027626	3.0358733	.957
	link	.7914127*	.1481138	<.001
	navigation	-2.4288733*	.4685182	<.001
	text	-.5264278*	.1455573	.011

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
	link	-2.417190	3.945662
	navigation	-5.934460	1.022360
	other	-3.194706	3.140353
	text	-3.733849	2.626640
input	alert	-15.736564	10.028504
	button	-6.126615	13.080749
	card	-7.775012	11.581552
	form	-8.613784	13.811829
	header	-6.177848	14.037727
	link	-4.915910	14.304261
	navigation	-8.238557	11.186335
	other	-5.702731	13.508257
	text	-6.233360	12.986029
link	alert	-16.145518	1.049107
	button	-1.647004	-.787212
	card	-4.064971	-1.516840
	form	-7.898697	3.708392
	header	-3.945662	2.417190
	input	-14.304261	4.915910
	navigation	-4.732143	-1.708429
	other	-1.260044	-.322781
	text	-1.865864	-.769817
navigation	alert	-13.039500	4.383662
	button	.532574	3.473781
	card	-1.468278	2.327039
	form	-4.846382	7.096649
	header	-1.022360	5.934460
	input	-11.186335	8.238557
	link	1.708429	4.732143
	other	.946483	3.911263
	text	.393076	3.411815
other	alert	-15.348972	1.835387
	button	-.736470	-.114922
	card	-3.238449	-.760536
	form	-7.099678	4.492198
	header	-3.140353	3.194706
	input	-13.508257	5.702731
	link	.322781	1.260044
	navigation	-3.911263	-.946483
	text	-.986971	-.065885

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
text	alert	-6.2303648	2.7170934	.393
	button	.1007322	.1330797	.999
	card	-1.4730649*	.4017426	.009
	form	-.7773121	1.8340400	1.000
	header	.5536044	1.0051351	1.000
	input	-3.3763348	3.0372009	.984
	link	1.3178405*	.1732061	<.001
	navigation	-1.9024455*	.4770450	.003
	other	.5264278*	.1455573	.011

Multiple Comparisons

Dependent Variable: contrast_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
text	alert	-14.827240	2.366511
	button	-.320332	.521796
	card	-2.744177	-.201952
	form	-6.580209	5.025585
	header	-2.626640	3.733849
	input	-12.986029	6.233360
	link	.769817	1.865864
	navigation	-3.411815	-.393076
	other	.065885	.986971

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

contrast_tritanopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05	
		1	2
link	2396	10.033475	
header	37	10.797711	
other	5614	10.824887	
button	11940	11.250583	11.250583
text	2519	11.351315	11.351315
form	11	12.128627	12.128627
card	251	12.824380	12.824380
navigation	173	13.253761	13.253761
input	4	14.727650	14.727650
alert	5		17.581680
Sig.		.407	.067

Means for groups in homogeneous subsets are displayed.

- Uses Harmonic Mean Sample Size = 17.278.
- The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Oneway

Notes

Output Created		20-APR-2026 18:48:37
Comments		
Input	Data	/Users/lukevassallo/Dev /6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.

Notes

Syntax	ONEWAY pass_protanopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).	
Resources	Processor Time	00:00:00.07
	Elapsed Time	00:00:00.00

ANOVA

pass_protanopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	41.666	9	4.630	37.527	<.001
Within Groups	2829.948	22940	.123		
Total	2871.613	22949			

ANOVA Effect Sizes^a

			95% Confidence Interval	
Point Estimate			Lower	Upper
pass_protanopia	Eta-squared	.015	.011	.017
	Epsilon-squared	.014	.011	.017
	Omega-squared Fixed-effect	.014	.011	.017
	Omega-squared Random-effect	.002	.001	.002

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	.107	.157	1.000
	card	.183	.159	.979
	form	.182	.189	.994
	header	.054	.167	1.000
	input	.000	.236	1.000
	link	.200	.157	.959
	navigation	.145	.159	.996
	other	.193	.157	.968
	text	.179	.157	.981
button	alert	-.107	.157	1.000
	card	.077*	.022	.022
	form	.075	.106	.999
	header	-.053	.058	.996
	input	-.107	.176	1.000
	link	.094*	.008	<.001
	navigation	.038	.027	.926
	other	.087*	.006	<.001
	text	.072*	.008	<.001
card	alert	-.183	.159	.979
	button	-.077*	.022	.022
	form	-.001	.108	1.000
	header	-.129	.062	.535
	input	-.183	.177	.990
	link	.017	.023	.999
	navigation	-.039	.035	.983
	other	.010	.023	1.000
	text	-.004	.023	1.000
form	alert	-.182	.189	.994
	button	-.075	.106	.999
	card	.001	.108	1.000
	header	-.128	.121	.988
	input	-.182	.205	.997
	link	.019	.106	1.000
	navigation	-.037	.109	1.000
	other	.011	.106	1.000
	text	-.003	.106	1.000
header	alert	-.054	.167	1.000
	button	.053	.058	.996
	card	.129	.062	.535
	form	.128	.121	.988
	input	-.054	.185	1.000

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
alert	button	-.39	.60
	card	-.32	.69
	form	-.42	.78
	header	-.48	.58
	input	-.75	.75
	link	-.30	.70
	navigation	-.36	.65
	other	-.30	.69
	text	-.32	.68
button	alert	-.60	.39
	card	.01	.15
	form	-.26	.41
	header	-.24	.13
	input	-.66	.45
	link	.07	.12
	navigation	-.05	.12
	other	.07	.10
	text	.05	.10
card	alert	-.69	.32
	button	-.15	-.01
	form	-.34	.34
	header	-.32	.07
	input	-.74	.38
	link	-.06	.09
	navigation	-.15	.07
	other	-.06	.08
	text	-.08	.07
form	alert	-.78	.42
	button	-.41	.26
	card	-.34	.34
	header	-.51	.25
	input	-.83	.47
	link	-.32	.35
	navigation	-.38	.31
	other	-.32	.35
	text	-.34	.33
header	alert	-.58	.48
	button	-.13	.24
	card	-.07	.32
	form	-.25	.51
	input	-.64	.53

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	link	.146	.058	.261
	navigation	.090	.064	.921
	other	.139	.058	.324
	text	.125	.058	.493
input	alert	.000	.236	1.000
	button	.107	.176	1.000
	card	.183	.177	.990
	form	.182	.205	.997
	header	.054	.185	1.000
	link	.200	.176	.981
	navigation	.145	.178	.998
	other	.193	.176	.985
	text	.179	.176	.991
link	alert	-.200	.157	.959
	button	-.094*	.008	<.001
	card	-.017	.023	.999
	form	-.019	.106	1.000
	header	-.146	.058	.261
	input	-.200	.176	.981
	navigation	-.056	.028	.586
	other	-.007	.009	.998
	text	-.021	.010	.510
navigation	alert	-.145	.159	.996
	button	-.038	.027	.926
	card	.039	.035	.983
	form	.037	.109	1.000
	header	-.090	.064	.921
	input	-.145	.178	.998
	link	.056	.028	.586
	other	.049	.027	.736
	text	.035	.028	.964
other	alert	-.193	.157	.968
	button	-.087*	.006	<.001
	card	-.010	.023	1.000
	form	-.011	.106	1.000
	header	-.139	.058	.324
	input	-.193	.176	.985
	link	.007	.009	.998
	navigation	-.049	.027	.736
	text	-.014	.008	.802

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
	link	-.04	.33
	navigation	-.11	.29
	other	-.04	.32
	text	-.06	.31
input	alert	-.75	.75
	button	-.45	.66
	card	-.38	.74
	form	-.47	.83
	header	-.53	.64
	link	-.36	.76
	navigation	-.42	.71
	other	-.36	.75
	text	-.38	.74
link	alert	-.70	.30
	button	-.12	-.07
	card	-.09	.06
	form	-.35	.32
	header	-.33	.04
	input	-.76	.36
	navigation	-.14	.03
	other	-.03	.02
	text	-.05	.01
navigation	alert	-.65	.36
	button	-.12	.05
	card	-.07	.15
	form	-.31	.38
	header	-.29	.11
	input	-.71	.42
	link	-.03	.14
	other	-.04	.13
	text	-.05	.12
other	alert	-.69	.30
	button	-.10	-.07
	card	-.08	.06
	form	-.35	.32
	header	-.32	.04
	input	-.75	.36
	link	-.02	.03
	navigation	-.13	.04
	text	-.04	.01

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
text	alert	-.179	.157	.981
	button	-.072*	.008	<.001
	card	.004	.023	1.000
	form	.003	.106	1.000
	header	-.125	.058	.493
	input	-.179	.176	.991
	link	.021	.010	.510
	navigation	-.035	.028	.964
	other	.014	.008	.802

Multiple Comparisons

Dependent Variable: pass_protanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
text	alert	-.68	.32
	button	-.10	-.05
	card	-.07	.08
	form	-.33	.34
	header	-.31	.06
	input	-.74	.38
	link	-.01	.05
	navigation	-.12	.05
	other	-.01	.04

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

pass_protanopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05 1
link	2396	.80
other	5614	.81
card	251	.82
form	11	.82
text	2519	.82
navigation	173	.86
button	11940	.89
header	37	.95
alert	5	1.00
input	4	1.00
Sig.		.809

Means for groups in homogeneous subsets are displayed.

- a. Uses Harmonic Mean Sample Size = 17.278.
- b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Oneway

Notes

Output Created		20-APR-2026 18:49:13
Comments		
Input	Data	/Users/lukevassallo/Dev/6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY pass_deuteranopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL (0.95) /POSTHOC=TUKEY ALPHA(0.05).
Resources	Processor Time	00:00:00.06
	Elapsed Time	00:00:00.00

ANOVA

pass_deuteranopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	32.895	9	3.655	28.645	<.001
Within Groups	2927.077	22940	.128		
Total	2959.973	22949			

ANOVA Effect Sizes^a

			95% Confidence Interval	
		Point Estimate	Lower	Upper
pass_deuteranopia	Eta-squared	.011	.008	.014
	Epsilon-squared	.011	.008	.013
	Omega-squared Fixed-effect	.011	.008	.013
	Omega-squared Random-effect	.001	.001	.001

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: pass_deuteranopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	.117	.160	.999
	card	.183	.161	.981
	form	.182	.193	.995
	header	.054	.170	1.000
	input	.000	.240	1.000
	link	.199	.160	.965
	navigation	.145	.162	.997
	other	.193	.160	.971
	text	.183	.160	.980
button	alert	-.117	.160	.999
	card	.067	.023	.100
	form	.065	.108	1.000
	header	-.063	.059	.988
	input	-.117	.179	1.000
	link	.082 [*]	.008	<.001
	navigation	.028	.027	.991
	other	.077 [*]	.006	<.001
	text	.066 [*]	.008	<.001
card	alert	-.183	.161	.981
	button	-.067	.023	.100
	form	-.001	.110	1.000
	header	-.129	.063	.560
	input	-.183	.180	.991
	link	.016	.024	1.000
	navigation	-.039	.035	.985
	other	.010	.023	1.000
	text	-.001	.024	1.000

Multiple Comparisons

Dependent Variable: pass_deutanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
alert	button	-.39	.62
	card	-.33	.69
	form	-.43	.79
	header	-.48	.59
	input	-.76	.76
	link	-.31	.71
	navigation	-.37	.66
	other	-.31	.70
	text	-.32	.69
button	alert	-.62	.39
	card	-.01	.14
	form	-.28	.41
	header	-.25	.12
	input	-.68	.45
	link	.06	.11
	navigation	-.06	.11
	other	.06	.09
	text	.04	.09
card	alert	-.69	.33
	button	-.14	.01
	form	-.35	.35
	header	-.33	.07
	input	-.75	.39
	link	-.06	.09
	navigation	-.15	.07
	other	-.06	.08
	text	-.08	.07

Multiple Comparisons

Dependent Variable: pass_deutanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
form	alert	-.182	.193	.995
	button	-.065	.108	1.000
	card	.001	.110	1.000
	header	-.128	.123	.990
	input	-.182	.209	.997
	link	.017	.108	1.000
	navigation	-.037	.111	1.000
	other	.011	.108	1.000
	text	.001	.108	1.000
header	alert	-.054	.170	1.000
	button	.063	.059	.988
	card	.129	.063	.560
	form	.128	.123	.990
	input	-.054	.188	1.000
	link	.145	.059	.296
	navigation	.090	.065	.928
	other	.139	.059	.349
	text	.129	.059	.475
input	alert	.000	.240	1.000
	button	.117	.179	1.000
	card	.183	.180	.991
	form	.182	.209	.997
	header	.054	.188	1.000
	link	.199	.179	.983
	navigation	.145	.181	.999
	other	.193	.179	.987
	text	.183	.179	.991
link	alert	-.199	.160	.965
	button	-.082 [*]	.008	<.001
	card	-.016	.024	1.000
	form	-.017	.108	1.000
	header	-.145	.059	.296
	input	-.199	.179	.983
	navigation	-.055	.028	.641
	other	-.006	.009	1.000
	text	-.016	.010	.841
navigation	alert	-.145	.162	.997
	button	-.028	.027	.991
	card	.039	.035	.985
	form	.037	.111	1.000
	header	-.090	.065	.928
	input	-.145	.181	.999

Multiple Comparisons

Dependent Variable: pass_deutanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
form	alert	-.79	.43
	button	-.41	.28
	card	-.35	.35
	header	-.52	.26
	input	-.84	.48
	link	-.32	.36
	navigation	-.39	.31
	other	-.33	.35
	text	-.34	.34
header	alert	-.59	.48
	button	-.12	.25
	card	-.07	.33
	form	-.26	.52
	input	-.65	.54
	link	-.04	.33
	navigation	-.11	.30
	other	-.05	.33
	text	-.06	.32
input	alert	-.76	.76
	button	-.45	.68
	card	-.39	.75
	form	-.48	.84
	header	-.54	.65
	link	-.37	.76
	navigation	-.43	.72
	other	-.37	.76
	text	-.38	.75
link	alert	-.71	.31
	button	-.11	-.06
	card	-.09	.06
	form	-.36	.32
	header	-.33	.04
	input	-.76	.37
	navigation	-.14	.03
	other	-.03	.02
	text	-.05	.02
navigation	alert	-.66	.37
	button	-.11	.06
	card	-.07	.15
	form	-.31	.39
	header	-.30	.11
	input	-.72	.43

Multiple Comparisons

Dependent Variable: pass_deutanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	link	.055	.028	.641
	other	.049	.028	.755
	text	.038	.028	.940
other	alert	-.193	.160	.971
	button	-.077*	.006	<.001
	card	-.010	.023	1.000
	form	-.011	.108	1.000
	header	-.139	.059	.349
	input	-.193	.179	.987
	link	.006	.009	1.000
	navigation	-.049	.028	.755
	text	-.011	.009	.965
text	alert	-.183	.160	.980
	button	-.066*	.008	<.001
	card	.001	.024	1.000
	form	-.001	.108	1.000
	header	-.129	.059	.475
	input	-.183	.179	.991
	link	.016	.010	.841
	navigation	-.038	.028	.940
	other	.011	.009	.965

Multiple Comparisons

Dependent Variable: pass_deutanopia

Tukey HSD

		95% Confidence Interval	
(I) component_type_num	(J) component_type_num	Lower Bound	Upper Bound
	link	-.03	.14
	other	-.04	.14
	text	-.05	.13
other	alert	-.70	.31
	button	-.09	-.06
	card	-.08	.06
	form	-.35	.33
	header	-.33	.05
	input	-.76	.37
	link	-.02	.03
	navigation	-.14	.04
	text	-.04	.02
text	alert	-.69	.32
	button	-.09	-.04
	card	-.07	.08
	form	-.34	.34
	header	-.32	.06
	input	-.75	.38
	link	-.02	.05
	navigation	-.13	.05
	other	-.02	.04

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

pass_deuteranopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05 1
link	2396	.80
other	5614	.81
card	251	.82
text	2519	.82
form	11	.82
navigation	173	.86
button	11940	.88
header	37	.95
alert	5	1.00
input	4	1.00
Sig.		.829

Means for groups in homogeneous subsets are displayed.

- a. Uses Harmonic Mean Sample Size = 17.278.
- b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

Oneway

Notes

Output Created		20-APR-2026 18:49:23
Comments		
Input	Data	/Users/lukevassallo/Dev/6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing.
	Cases Used	Statistics for each analysis are based on cases with no missing data for any variable in the analysis.
Syntax		ONEWAY pass_tritanopia BY component_type_num /ES=OVERALL /MISSING ANALYSIS /CRITERIA=CILEVEL(0.95) /POSTHOC=TUKEY ALPHA(0.05).
Resources	Processor Time	00:00:00.07
	Elapsed Time	00:00:00.00

ANOVA

pass_tritanopia

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	38.893	9	4.321	34.927	<.001
Within Groups	2838.371	22940	.124		
Total	2877.265	22949			

ANOVA Effect Sizes^a

			95% Confidence Interval	
Point Estimate			Lower	Upper
pass_tritanopia	Eta-squared	.014	.010	.016
	Epsilon-squared	.013	.010	.016
	Omega-squared Fixed-effect	.013	.010	.016
	Omega-squared Random-effect	.001	.001	.002

a. Eta-squared and Epsilon-squared are estimated based on the fixed-effect model.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
alert	button	.108	.157	1.000
	card	.183	.159	.979
	form	.182	.190	.994
	header	.054	.168	1.000
	input	.000	.236	1.000
	link	.198	.157	.963
	navigation	.145	.160	.996
	other	.193	.157	.968
	text	.177	.157	.982
button	alert	-.108	.157	1.000
	card	.075 [*]	.022	.029
	form	.073	.106	1.000
	header	-.054	.058	.995
	input	-.108	.176	1.000
	link	.089 [*]	.008	<.001
	navigation	.036	.027	.945
	other	.084 [*]	.006	<.001
	text	.069 [*]	.008	<.001
card	alert	-.183	.159	.979
	button	-.075 [*]	.022	.029
	form	-.001	.108	1.000
	header	-.129	.062	.537
	input	-.183	.177	.990
	link	.015	.023	1.000
	navigation	-.039	.035	.983
	other	.010	.023	1.000
	text	-.006	.023	1.000

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
alert	button	-.39	.61
	card	-.32	.69
	form	-.42	.78
	header	-.48	.58
	input	-.75	.75
	link	-.30	.70
	navigation	-.36	.65
	other	-.31	.69
	text	-.32	.68
button	alert	-.61	.39
	card	.00	.15
	form	-.26	.41
	header	-.24	.13
	input	-.67	.45
	link	.06	.11
	navigation	-.05	.12
	other	.07	.10
	text	.04	.09
card	alert	-.69	.32
	button	-.15	.00
	form	-.34	.34
	header	-.33	.07
	input	-.74	.38
	link	-.06	.09
	navigation	-.15	.07
	other	-.06	.08
	text	-.08	.07

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
form	alert	-.182	.190	.994
	button	-.073	.106	1.000
	card	.001	.108	1.000
	header	-.128	.121	.989
	input	-.182	.205	.997
	link	.016	.106	1.000
	navigation	-.037	.109	1.000
	other	.011	.106	1.000
	text	-.005	.106	1.000
header	alert	-.054	.168	1.000
	button	.054	.058	.995
	card	.129	.062	.537
	form	.128	.121	.989
	input	-.054	.185	1.000
	link	.144	.058	.286
	navigation	.090	.064	.921
	other	.139	.058	.330
	text	.123	.058	.519
input	alert	.000	.236	1.000
	button	.108	.176	1.000
	card	.183	.177	.990
	form	.182	.205	.997
	header	.054	.185	1.000
	link	.198	.176	.982
	navigation	.145	.178	.998
	other	.193	.176	.985
	text	.177	.176	.992
link	alert	-.198	.157	.963
	button	-.089 [*]	.008	<.001
	card	-.015	.023	1.000
	form	-.016	.106	1.000
	header	-.144	.058	.286
	input	-.198	.176	.982
	navigation	-.053	.028	.652
	other	-.005	.009	1.000
	text	-.021	.010	.549
navigation	alert	-.145	.160	.996
	button	-.036	.027	.945
	card	.039	.035	.983
	form	.037	.109	1.000
	header	-.090	.064	.921
	input	-.145	.178	.998

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
form	alert	-.78	.42
	button	-.41	.26
	card	-.34	.34
	header	-.51	.25
	input	-.83	.47
	link	-.32	.35
	navigation	-.38	.31
	other	-.32	.35
	text	-.34	.33
header	alert	-.58	.48
	button	-.13	.24
	card	-.07	.33
	form	-.25	.51
	input	-.64	.53
	link	-.04	.33
	navigation	-.11	.29
	other	-.04	.32
	text	-.06	.31
input	alert	-.75	.75
	button	-.45	.67
	card	-.38	.74
	form	-.47	.83
	header	-.53	.64
	link	-.36	.75
	navigation	-.42	.71
	other	-.36	.75
	text	-.38	.73
link	alert	-.70	.30
	button	-.11	-.06
	card	-.09	.06
	form	-.35	.32
	header	-.33	.04
	input	-.75	.36
	navigation	-.14	.03
	other	-.03	.02
	text	-.05	.01
navigation	alert	-.65	.36
	button	-.12	.05
	card	-.07	.15
	form	-.31	.38
	header	-.29	.11
	input	-.71	.42

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	Mean Difference (I-J)	Std. Error	Sig.
	link	.053	.028	.652
	other	.048	.027	.746
	text	.033	.028	.976
other	alert	-.193	.157	.968
	button	-.084 [*]	.006	<.001
	card	-.010	.023	1.000
	form	-.011	.106	1.000
	header	-.139	.058	.330
	input	-.193	.176	.985
	link	.005	.009	1.000
	navigation	-.048	.027	.746
	text	-.016	.008	.683
text	alert	-.177	.157	.982
	button	-.069 [*]	.008	<.001
	card	.006	.023	1.000
	form	.005	.106	1.000
	header	-.123	.058	.519
	input	-.177	.176	.992
	link	.021	.010	.549
	navigation	-.033	.028	.976
	other	.016	.008	.683

Multiple Comparisons

Dependent Variable: pass_tritanopia

Tukey HSD

(I) component_type_num	(J) component_type_num	95% Confidence Interval	
		Lower Bound	Upper Bound
	link	-.03	.14
	other	-.04	.13
	text	-.05	.12
other	alert	-.69	.31
	button	-.10	-.07
	card	-.08	.06
	form	-.35	.32
	header	-.32	.04
	input	-.75	.36
	link	-.02	.03
	navigation	-.13	.04
	text	-.04	.01
	alert	-.68	.32
	button	-.09	-.04
	card	-.07	.08
text	form	-.33	.34
	header	-.31	.06
	input	-.73	.38
	link	-.01	.05
	navigation	-.12	.05
	other	-.01	.04

*. The mean difference is significant at the 0.05 level.

Homogeneous Subsets

pass_tritanopia

Tukey HSD^{a,b}

component_type_num	N	Subset for alpha = 0.05 1
link	2396	.80
other	5614	.81
card	251	.82
form	11	.82
text	2519	.82
navigation	173	.86
button	11940	.89
header	37	.95
alert	5	1.00
input	4	1.00
Sig.		.822

Means for groups in homogeneous subsets are displayed.

- a. Uses Harmonic Mean Sample Size = 17.278.
- b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.

GGraph

Notes

Output Created		20-APR-2026 19:03:34
Comments		
Input	Data	/Users/lukevassallo/Dev/6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950

Notes

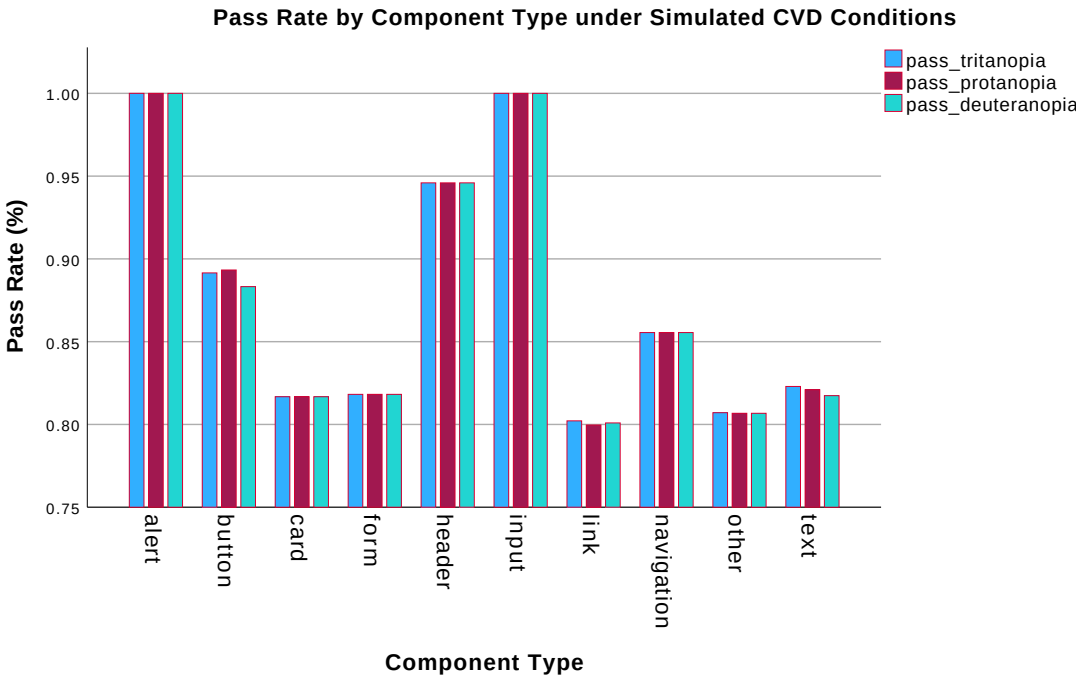
Syntax

```
GGRAPH
  /GRAPHDATASET
  NAME="graphdataset"
  VARIABLES=component_t
  ype_num MEAN
  (pass_tritanopia)
  MEAN
  (pass_protanopia) MEAN
  (pass_deutanopia)
  MISSING=LISTWISE
  REPORTMISSING=NO

  TRANSFORM=VARSTOCA
  SES(SUMMARY="
  #SUMMARY" INDEX="
  #INDEX")
  /GRAPHSPEC
  SOURCE=INLINE.
  BEGIN GPL
    SOURCE: s=userSource
    (id("graphdataset"))
    DATA:
    component_type_num=c
    ol(source(s), name
    ("component_type_num")
    , unit.category())
    DATA: SUMMARY=col
    (source(s), name
    ("SUMMARY"))
    DATA: INDEX=col
    (source(s), name
    ("INDEX"), unit.
    category())
    COORD: rect(dim(1,2),
    cluster(3,0))
    GUIDE: axis(dim(3),
    label
    ("component_type_num")
    )
    GUIDE: axis(dim(2),
    label("Mean"))
    GUIDE: legend
    (aesthetic(aesthetic.
    color.interior), label(""))
    GUIDE: text.title(label
    ("Simple Bar Mean of
    pass_tritanopia, Mean of
    pass_protanopia, Mean
    of ",
    "pass_deutanopia
    by
    component_type_num..."
    ))
    SCALE: cat(dim(3),
    include("1", "2", "3", "4",
    "5", "6", "7", "8", "9",
    "10"))
    SCALE: linear(dim(2),
    include(0))
    SCALE: cat(aesthetic
    (aesthetic.color.interior),
    include(
    "0", "1", "2"))
    SCALE: cat(dim(1),
    include(
    "0", "1", "2"))
    ELEMENT: interval
    (position
    (INDEX*SUMMARY*comp
    onent_type_num), color.
    interior(INDEX),
    shape.interior(shape.
    square))
  END GPL.
```

Notes

Resources	Processor Time	00:00:00.32
	Elapsed Time	00:00:00.00



GGraph

Notes

Output Created		20-APR-2026 19:05:54
Comments		
Input	Data	/Users/lukevassallo/Dev/6.3A/Semester 2/Research Design/ResearchDesign2 LVCCD/output/detailed_results.csv
	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	22950

Notes

Syntax

```
GGRAPH
/GRAPHDATASET
NAME="graphdataset"
VARIABLES=component_t
ype_num MEAN
(contrast_protanopia)
MEAN
(contrast_tritanopia)
MEAN
(contrast_deutanopia)
MISSING=LISTWISE
REPORTMISSING=NO

TRANSFORM=VARSTOCA
SES(SUMMARY="
#SUMMARY" INDEX="
#INDEX")
/GRAPHSPEC
SOURCE=INLINE.
BEGIN GPL
SOURCE: s=userSource
(id("graphdataset"))
DATA:
component_type_num=c
ol(source(s), name
("component_type_num")
, unit.category())
DATA: SUMMARY=col
(source(s), name
("#SUMMARY"))
DATA: INDEX=col
(source(s), name
("#INDEX"), unit.
category())
COORD: rect(dim(1,2),
cluster(3,0))
GUIDE: axis(dim(3),
label
("component_type_num")
)
GUIDE: axis(dim(2),
label("Mean"))
GUIDE: legend
(aesthetic(aesthetic.
color.interior), label(""))
GUIDE: text.title(label
("Simple Bar Mean of
contrast_protanopia,
Mean of
contrast_tritanopia, ",
"Mean of
contrast_deutanopia
by
component_type_num..."
))
SCALE: cat(dim(3),
include("1", "2", "3", "4",
"5", "6", "7", "8", "9",
"10"))
SCALE: linear(dim(2),
include(0))
SCALE: cat(aesthetic
(aesthetic.color.interior),
include(
"0", "1", "2"))
SCALE: cat(dim(1),
include(
"0", "1", "2"))
ELEMENT: interval
(position
(INDEX*SUMMARY*comp
onent_type_num), color.
interior(INDEX),
shape.interior(shape.
square))
END GPL.
```

Notes

Resources	Processor Time	00:00:00.22
	Elapsed Time	00:00:00.00

